0

master run latch for the parts sorting simulation. start seals in the machine run bit, and Stop drops the system out of run

stop pushbutton
PB_stop

start push button
PB_Start

master run bit
System_Run

master run bit
System_Run
L

1

Turns on the conveyor whenever the system is in run mode

master run bit
System_Run

Simulated conveyor motor run
Conveyor_Run
U

2

generates a repeating time base used to simulate conveyor movement. Each timer completion represents one conveyor index pulse

Simulated conveyor motor run
Conveyor_Run

Conveyor pulse generator

TON		
Timer	T_Pulse	EN
Preset	500	DN
Accum	0	

3

Creates a one-scan shift pulse each time the conveyor pulse timer completes

Conveyor pulse generator
T_Pulse.DN

Storage bit for conveyor pulse one-shot
ONS_Shift
ONS

One-shot conveyor movement pulse
Shift_Pulse

4

Resets the coneyor pulse timer after each completed pulse so the timer can continously generate new shift events while the conveyor is running

Conveyor pulse generator
T_Pulse.DN

Conveyor pulse generator
T_Pulse
RES

5

creates a one-scan pulse when a new part is detected at the entry photoeye so the part is loaded only once into the tracking registers

entry photoeye detects a new part
PE_Entry

storage bit for entry sensor one-shot
ONS_Enrty
ONS

One-shot pulse for new part entry
Entry_Event

6

counts every new part entering the conveyor system

One-shot pulse for new part entry
Entry_Event

counts all parts entering

CTU		
Counter	C_TotalParts	CU
Preset	9999	DN
Accum	0	

7

Counts parts classified as good when a part enters and the simulated vision result is not bad

One-shot pulse for new part entry
Entry_Event

simulated vision result, 1 = bad, 0 = good
Vision_Bad

counts all good parts

CTU		
Counter	C_GoodParts	CU
Preset	9999	DN
Accum	0	

8

counts parts classified as bad when a part enters and the simulated vision result indicates a reject condition

One-shot pulse for new part entry
Entry_Event

simulated vision result, 1 = bad, 0 = good
Vision_Bad

CTU
Counter C_BadParts
Preset 9999
Accum 0
CU
DN

9

Loads a 1 into the bad-part shift register when the incoming part is classified as bad. Good parts load a 0 by leaving this bit off

One-shot pulse for new part entry
Entry_Event

simulated vision result, 1 = bad, 0 = good
Vision_Bad

Bit loaded into bad-part register
Load_Badbit

10

Loads a 1 into the part-present shift register whenever any part enters the system, regardless of classification

One-shot pulse for new part entry
Entry_Event

Bit loaded into part-present register
Load_PresentBit

11

Shifts the bad-part classification register one position on each movement pulse. A bad part enters the register as 1 and a good part enters as 0

One-shot conveyor movement pulse
Shift_Pulse

Control structure for bad-part shift register

BSL
Array BadPartTrack
Control BSL_Bad_Control
Source Bit Load_Badbit
Length 10
EN
DN

12

Shifts the part-present register one position on each conveyor movement pulse so the plc knows whether a physical part exists at each tracked position.

One-shot conveyor movement pulse
Shift_Pulse

One-scan manual pulse
Manual_Step_Pulse

One-shot conveyor movement pulse
Shift_Pulse

Control structure for part-present shift register

BSL
Array PartPresentTrack
Control BSL_Present_Control
Source Bit Load_PresentBit
Length 10
EN
DN

13

Indicates that a tracked part has reached the reject station position in the conveyor model.

Tracks physical part location
PartPresentTrack[0].9

indicates tracked part at reject station
Part_At_Reject

14

Indicates that the tracked part currently at the reject station is classified as bad

tracks bad part
BadPartTrack[0].9

Indicates tracked bad bit at reject station
Bad_At_Reject

15

Creates an internal request only when a part is physically present at the reject station and its tracked classification is bad

indicates tracked part at reject station
Part_At_Reject

Indicates tracked bad bit at reject station
Bad_At_Reject

Internal request to fire reject gate
Reject_Request

16

starts the reject gate pulse timer whenever a bad part reaches the reject station. The timer determines how long the diverter remains energized.

Internal request to fire reject gate
Reject_Request

Reject gate pulse timer

TON		
Timer	T_Reject	EN
Preset	300	DN
Accum	0	

17

Energizes the reject gate during the active reject window and turns it off once the reject timer is complete.

reject/diverter actuator output
Reject_Gate

Reject gate pulse timer
T_Reject.DN

reject/diverter actuator output
Reject_Gate

18

Counts each bad part rejected at the diverter station one time per reject event

Internal request to fire reject gate
Reject_Request

Storage bit for rejected counter one-shot
ONS_RejectCount
ONS

counts rejected bad parts

CTU		
Counter	C_RejectedParts	CU
Preset	0	DN
Accum	0	

19

Resets all production and reject counters when the system reset pushbutton is pressed

system reset pushbutton
PB_Reset

counts all parts entering
C_TotalParts
RES

counts all good parts
C_GoodParts
RES

C_BadParts
RES

counts rejected bad parts
C_RejectedParts
RES

20

clears all tracked part data from the shift register arrays so the conveyor model returns to an empty state

system reset pushbutton
PB_Reset

Conveyor pulse generator
T_Pulse
RES

Reject gate pulse timer
T_Reject
RES

21

Clears all tracked part data from the shift register arrays so the conveyor model returns to an empty state

system reset pushbutton
PB_Reset

tracks bad part
CLR
Dest BadPartTrack[0]
0

Tracks physical part location
CLR
Dest PartPresentTrack[0]
0

22

returns the machine to a stopped state after reset so the sorting process restarts from a known safe condition

system reset pushbutton
PB_Reset

master run bit
System_Run
U

23

Manual shift pushbutton
PB_step

One-shot storage
ONS_Step
ONS

One-scan manual pulse
Manual_Step_Pulse

(End)